ASA and AAS Triangle Congruence

Geometry Worksheet · Grade 8–10

Name: _____ Date: _____

Learning Objectives

- Identify and apply the Angle-Side-Angle (ASA) postulate to prove triangle congruence
- Identify and apply the Angle-Angle-Side (AAS) postulate to prove triangle congruence
- Distinguish between included and non-included sides in the context of triangle congruence

Problems

1. State the ASA Postulate in your own words: what three parts of two triangles must be congruent?

2. In triangle ABC and triangle DEF, if ∠A ≅ ∠D, AC ≅ DF, and ∠C ≅ ∠F, which postulate proves the triangles congruent?

$$\angle A \cong \angle D,\ AC \cong DF,\ \angle C \cong \angle F$$

3. In triangle ABC and triangle DEF, if ∠A ≅ ∠D, ∠B ≅ ∠E, and AC ≅ DF, which postulate proves the triangles congruent?

$$\angle A \cong \angle D,\ \angle B \cong \angle E,\ AC \cong DF$$

4. Explain the difference between an included side and a non-included side as used in the ASA and AAS postulates.

5. Four triangles are labeled A, B, C, and D. Triangles A and C each show the pattern angle-angle-side (AAS), while triangles B and D each show angle-side-angle (ASA). Which pairs of triangles are congruent by ASA?

6. Given that DB bisects ∠ABC, identify the two congruent angles this creates and write a congruence statement for them.

$$\angle 1 \cong \angle 2$$

7. In a two-column proof showing $\Delta ABD \cong \Delta CBD$ by ASA, list the three statements (in order) that form the 'if' part of the proof, given that DB bisects $\angle ABC$ and $\angle 3 \cong \angle 4$.

$$\angle 1 \cong \angle 2,\ DB \cong DB,\ \angle 3 \cong \angle 4$$

8. Determine whether the following information is sufficient to prove triangle congruence by ASA or AAS, or neither: $\angle P \cong \angle X$, $PQ \cong XY$, $\angle R \cong \angle Z$ in triangles PQR and XYZ.

$$\angle P \cong \angle X,\ PQ \cong XY,\ \angle R \cong \angle Z$$

9. Write a complete two-column proof to show $\Delta ABD \cong \Delta CBD$ given: DB bisects $\angle ABC$ (so $\angle 1 \cong \angle 2$) and $\angle 3 \cong \angle 4$, with DB as the shared side.

$$\triangle ABD \cong \triangle CBD$$

10. Triangle MNP and triangle QRS have $\angle M \cong \angle Q$ and $\angle N \cong \angle R$. If you are told that $NP \cong RS$, prove or disprove that the triangles are congruent, and if so, state which postulate applies.

$$\angle M \cong \angle Q,\ \angle N \cong \angle R,\ NP \cong RS$$

ASA and AAS Triangle Congruence — Answer Key

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Answer Key

1. Answer: Two angles and the included side of one triangle must be congruent to two angles and the included side of the other triangle.

- Recall that ASA stands for Angle-Side-Angle.
- The side must be between (included by) the two congruent angles.

2. Answer: ASA Postulate

- Identify the pattern: angle, then side AC (between ∠A and ∠C), then angle — this is Angle-Side-Angle.
- Since the side AC is included between the two congruent angles, the ASA Postulate applies.

3. Answer: AAS Postulate

- Side AC is not between ∠A and ∠B (∠B is at vertex B, which is not an endpoint of AC), making AC a non-included side.
- Two angles and a non-included side match the AAS Postulate.

4. Answer: An included side lies directly between the two angles being used; a non-included side does not connect both of those angles.

- An included side shares both endpoints with the two angles in question.
- A non-included side has at least one endpoint that is not one of the two angles being compared.

5. Answer: Triangles B and D are congruent by ASA.

- Identify the marking pattern for each triangle: B and D both show angle-side-angle.
- Because B and D share the same ASA pattern with corresponding congruent parts, they are congruent by ASA.

6. Answer: ∠1 ≅ ∠2 (the bisector divides ∠ABC into two equal halves)

- A bisector divides an angle into two congruent parts.
- Therefore ∠ABD (∠1) ≅ ∠DBC (∠2) by the definition of an angle bisector.

7. Answer: ∠1 ≅ ∠2 (given/def. of bisector), DB ≅ DB (Reflexive Property), ∠3 ≅ ∠4 (given)

- Statement 1: ∠1 ≅ ∠2 — Reason: DB bisects ∠ABC (given), definition of angle bisector.
- Statement 2: DB ≅ DB — Reason: Reflexive Property of Congruence.
- Statement 3: ∠3 ≅ ∠4 — Reason: Given.

8. Answer: AAS Postulate (side PQ is non-included since ∠R is not at vertex Q relative to the pair ∠P and ∠R)

- Check if the side PQ lies between the two congruent angles ∠P and ∠R: PQ connects vertices P and Q, but ∠R is at vertex R — not an endpoint of PQ.
- Since the side is non-included relative to ∠P and ∠R, this matches the AAS Postulate.

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9. Answer: ΔABD ≅ ΔCBD by ASA Postulate

- Statement 1: ∠1 ≅ ∠2 | Reason: DB bisects ∠ABC; Definition of Angle Bisector.
- Statement 2: DB ≅ DB | Reason: Reflexive Property of Congruence.
- Statement 3: ∠3 ≅ ∠4 | Reason: Given.
- Statement 4: ΔABD ≅ ΔCBD | Reason: ASA Postulate (∠-side-∠ pattern confirmed).

10. Answer: Yes, ΔMNP ≅ ΔQRS by AAS Postulate

- Identify the two pairs of congruent angles: ∠M ≅ ∠Q and ∠N ≅ ∠R.
- Check whether NP is the included side between ∠N and ∠P — since ∠M (not ∠P) is used instead, NP is a non-included side with respect to ∠M and ∠N.
- Two congruent angles and a non-included corresponding side satisfy the AAS Postulate, so ΔMNP ≅ ΔQRS by AAS.

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